Mathematical Analysis
Foundamental Theorems on Real Number Theory

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Real Number System

There are at least two approaches to defining the real number system.

Approach 1. Start with the set of natural numbers $\mathbb{N}$, construct the set $\mathbb{Z}$ of all integers $(\mathbb{N} \cup \{0\} \cup (-\mathbb{N}))$, and next, the set $\mathbb{Q}$ of rationals and then the set of $\mathbb{R}$ of real numbers either as the set of all Cauchy sequences of rationals or as Dedekind cuts. The step going from $\mathbb{Q}$ to $\mathbb{R}$ via Cauchy sequences is also available for completing any incomplete metric space.

Real Number System

There are at least two approaches to defining the real number system.

Approach 2. Define the set of real numbers, $\mathbb{R}$, as a set that satisfies three sets of axioms. The first set is algebraic involving addition and multiplication. The second set is on ordering that, with the first, makes $\mathbb{R}$ an ordered field (Royden 1988). The third set is a single axiom known as the completeness axiom. Thus $\mathbb{R}$ is defined as complete ordered field.

Additive Group

A group is a set, G, together with an operation $+$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 + a_2$. To qualify as a group, the set and operation, $(G, +)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 + a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(3) Identity element: There exists an element 0 in G such that, for every element a in G, the equation $0 + a = a + 0 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Additive Inverse: For each a in G, there exists an inverse element $-a$ in G such that $a + (-a) = (-a) + a = 0$.

A group $(G, +)$ with the property that $a_1 + a_2 = a_2 + a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

Multiplicative Group

A group is a set, G, together with an operation $\times$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \times a_2$. To qualify as a group, the set and operation, $(G, \times)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \times a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(3) Identity element: There exists an element 1 in G such that, for every element a in G, the equation $1 \times a = a \times 1 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Multiplicative Inverse: For each a in G, there exists an inverse element a^{-1} in G such that $a \times a^{-1} = a^{-1} \times a = 1$.

A group $(G, \times)$ with the property that $a_1 \times a_2 = a_2 \times a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

Group

A group is a set, G, together with an operation $\cdot$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \cdot a_2$. To qualify as a group, the set and operation, $(G, \cdot)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \cdot a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \cdot a_2) \cdot a_3 = a_1 \cdot (a_2 \cdot a_3)$.

(3) Identity element: There exists an element e in G such that, for every element a in G, the equation $e \cdot a = a \cdot e = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Inverse element: For each a in G, there exists an inverse element a^{-1} in G such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

A group $(G, \cdot)$ with the property that $a_1 \cdot a_2 = a_2 \cdot a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

Ring

A ring is a set R equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) R is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in R.

(1.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in R, there exists an inverse element $-a$ in R such that $a + (-a) = (-a) + a = 0$.

(2) R is a monoid under multiplication $\times$, meaning that:

(2.1) Closure: For all a_1 and a_2 in R, the result of the operation, $a_1 \times a_2$, is also in R.

(2.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

Abelian Ring = Commutative Ring

A ring is a set R equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) R is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in R.

(1.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in R, there exists an inverse element $-a$ in R such that $a + (-a) = (-a) + a = 0$.

(2) R is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in R.

(2.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

Field

A field is a set F equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) F is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in F.

(1.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in F such that, for every element a in F, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in F, there exists an inverse element $-a$ in F such that $a + (-a) = (-a) + a = 0$.

(2) F is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in F.

(2.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in F such that, for every element a in F, the equation $1 \times a = a \times 1 = a$ holds.

(2.4) Multiplicative Inverse: For each $a \neq 0$ in F, there exists an inverse element a^{-1} in F such that $a \times a^{-1} = a^{-1} \times a = 1$.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in F, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

Field

If there is an element 1 in R such that $1 \neq 0$ and $1 \times a = a \times 1 = a$ for each element a in R, we say that R is a ring with unity or identity. A commutative ring R with identity is called an integral domain if, for every a_1 and a_2 in R such that $a_1 \times a_2 = 0$, either $a_1 = 0$ or $a_2 = 0$. A division ring is a ring R, with an identity, in which every non-zero element in R is a unit; that is, for each a in R with $a \neq 0$, there exists a unique element a^{-1} such that $a^{-1} \times a = a \times a^{-1} = 1$. A commutative division ring is called a field.

Rings: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Fields: $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Linear Space = Vector Space

A vector space over a field F is a set V together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors v_1 and v_2 and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set V).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar a and any vector v and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set V).

Elements of V are commonly called vectors. Elements of F are commonly called scalars.

Linear Space = Vector Space

(1) Commutativity: For all v_1 and v_2 in F, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in V.

(2) Associativity: For all v_1, v_2 and v_3 in V, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in V such that, for every element v in V, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each v in V, there exists an inverse element $-v$ in V such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field F for Scalar Multiplication: For each v in V, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any a_1 and a_2 in F and each v in V, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any a in F and each v_1 and v_2 in V, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any a_1 and a_2 in F and each v in V, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

Order Axioms

Order Axioms There is a subset $\mathbb{P}$ of the set $\mathbb{R}$ of real numbers, called the set of strictly positive numbers such that:

(1) For any real number a in $\mathbb{R}$, exactly one of the following holds: $a = 0$ or $a \in \mathbb{P}$ or $-a \in \mathbb{P}$.

(2) If a_1 and a_2 in $\mathbb{P}$, $a_1 + a_2 \in \mathbb{P}$.

(3) If a_1 and a_2 in $\mathbb{P}$, $a_1 \cdot a_2 \in \mathbb{P}$.

The "Less Than" Relation We now define

- $a_1 < a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ $(a_2 - a_1 \in \mathbb{R}_{++})$
- $a_1 \leqslant a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ or $a_1 = a_2$ $(a_2 - a_1 \in \mathbb{R}_{+})$
- $a_1 > a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ $(a_1 - a_2 \in \mathbb{R}_{++})$
- $a_1 \geqslant a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ or $a_1 = a_2$ $(a_1 - a_2 \in \mathbb{R}_{+})$

$a_1 < a_2$ if and only if $a_2 > a_1$ and $a_1 \leqslant a_2$ if and only if $a_2 \geqslant a_1$.

$$\mathbb{P} = \mathbb{R}_+ \setminus \{0\} = \mathbb{R}_{++}$$

Ordered Field

Any field F, together with two operations $\oplus$ and $\otimes$ and two members $0_\oplus$ and $0_\otimes$ of F, and a subset P of F, which satisfies the corresponding versions of (1) - (2) - (3), is called an ordered field.

Example. Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields.

Completeness Axiom

Dedekind Completeness Axiom Suppose A and B are two (non-empty) sets of real numbers with the properties:

(1) if $a \in A$ and $b \in B$, then $a < b$.

(2) every real number is in either A or B (in symbols; $A \cup B = \mathbb{R}$).

Then there is a unique real number c such that:

(1) if $a < c$ then $a \in A$, and

(2) if $b > c$ then $b \in B$.

Note that every number strictly less than c belongs to A and every number strictly more than c belongs to B. Moreover, either $c \in A$ or $c \in B$ by (2). Hence if $c \in A$ then $A = (-\infty, c]$ and $B = (c, \infty)$; while if $c \in B$ then $A = (-\infty, c)$ and $B = [c, \infty)$.

The pair of sets $\{A, B\}$ is called a Dedekind Cut.

The intuitive idea of Dedekind Completeness Axiom is that there are no "holes" in the real numbers.

Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$. Then either there exists a maximum in A or there exists a minimum in A^c:

(1) If there exists a maximum c in A, then $A = (-\infty, c]$ and $A^c = (c, \infty)$.

(2) If there exists a minimum c in A^c, then $A = (-\infty, c)$ and $A^c = [c, \infty)$.

Supremum and Infimum

(1) Let S be a set of real numbers. u is an upper bound for S if $x \leqslant u$ for all $x \in S$. A set S is bounded above if it has an upper bound.

(2) b is the least upper bound (or l.u.b. or supremum or sup) for S if b is an upper bound, and moreover $b \leqslant u$ whenever u is any upper bound for S.

(1) Let S be a set of real numbers. d is an lower bound for S if $x \geqslant d$ for all $x \in S$. A set S is bounded below if it has a lower bound.

(2) l is the greatest lower bound (or g.l.b. or infimum or inf) for S if l is an lower bound, and moreover $l \geqslant d$ whenever d is any lower bound for S.

Supremum and Infimum Completeness Axiom

Theorem of Supremum (Least Upper Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded above. Then S has a supremum in $\mathbb{R}$.

Theorem of Infimum (Greatest Lower Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded below. Then S has a infimum in $\mathbb{R}$.

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for S so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then a would also be an upper bound for S, which contradicts the definition of A.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then c is not an upper bound for S and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for S, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. c is less than any upper bound for S.

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then A is bounded above (by any element of B). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of B is an upper bound for A, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise a would be an upper bound for A which is less than the least upper bound c. Hence $a \in A$.
- Next suppose $b > c$. Since c is an upper bound for A (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for S so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then b would also be a lower bound for S, which contradicts the definition of B.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then c is not a lower bound for S and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for S, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. c is more than any lower bound for S.

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then B is bounded below (by any element of A). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since c is a lower bound for B (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of A is a lower bound for B, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise b would be a lower bound for B which is less than the greatest lower bound c. Hence $b \in B$.

Proposition on Supremum and Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

Definition of Supremum

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever u is any upper bound for S.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from $(2')$ it follows that u is not an upper bound of S. That is (2): $b \leqslant u$ whenever u is any upper bound for S.

Definition of Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) $l \geqslant d$ whenever d is any lower bound for S.

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant l + \varepsilon$ for every $x \in S$, i.e. $l + \varepsilon$ is a lower bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $d > l$ then from $(2')$ it follows that d is not a lower bound of S. That is (2): $l \geqslant d$ whenever d is any lower bound for S.

Bounded Monotone Sequence

Definition. A sequence is bounded if the set of terms from the sequence is bounded.

Definition. A sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is

(1) increasing if $x_n \leqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(2) decreasing if $x_n \geqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(3) strictly increasing if $x_n < x_{n+1}$ for all $n \in \mathbb{N}$;

(4) strictly decreasing if $x_n > x_{n+1}$ for all $n \in \mathbb{N}$.

A sequence is monotone if it is either increasing or decreasing.

Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is decreasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded below, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \geqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N < x + \varepsilon$, as otherwise $x + \varepsilon$ would be a lower bound. Choose such $N = N(\varepsilon)$. Since $x_N < x + \varepsilon$, then $x_n < x + \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x \leqslant x_n < x + \varepsilon$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = \sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\}$ and $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

$\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals if $I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$.

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = \sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by u_1. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set A and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by d_1. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $I_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $I_{n'}$ is a lower bound to the set B and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = \inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Proof We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$. Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that x is such that $d_n \leqslant x < D$. Since D is the supremum of the set A then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that x is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since U is the infimum of the set B then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

Proof The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by u_1, then it has a limit $D = \lim_{n\to\infty} d_n$.

The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by d_1, then it has a limit $U = \lim_{n\to\infty} u_n$.

Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k \to \infty} x_{n_k} = a$.

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that E is a bounded infinite set in $\mathbb{R}$, there exists a constant M such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide R_0 into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_1 with length M. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide I_N into two small closed intervals, then at least one small closed interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_{N+1} with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this x is one of accumulation points of E.
Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer n_0, $\frac{M}{2^{n_0}} < \delta$. The length of I_{n_0+1} is strictly less than δ, then $I_{n_0+1} \subseteq N(x, \delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x, \delta)$. x is an accumulation point of E.

Cauchy Convergence Criterion

Definition. $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is a Cauchy sequence if for any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Proof [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer K such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.
Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$.
Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim_{n\to\infty} x_n = a$.

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since y_{n+1} is the infimum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since y_{n+1} is the supremum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then $y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n\to\infty} x_n = a$.

Heine - Borel Theorem on Finite Coverings

A collection $\mathscr{H}$ of open sets is an open covering of a set S if every point in S is contained in a set H belonging to $\mathscr{H}$; that is, $S \subseteq \{H \in \mathscr{H}\}$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof (Reduction to Absurdity) Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $a \in H_0$. Note that H_0 is an open set, then there exists $\delta > 0$ such that $(x-\delta, x+\delta) \subseteq H_0$.
Note $\lim_{n\to\infty} D_n = a = \lim_{n\to\infty} U_n$, then there exists an integer N such that, when $n > N$, $D_n \in (x-\delta, x+\delta)$ and $U_n \in (x-\delta, x+\delta)$.

$$[D_n, U_n] \subseteq (x-\delta, x+\delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

Real Number Space $\mathbb{R}$

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$, then either there exists a maximum in A or there exists a minimum in A^c.

Theorem of Supremum and Infimum Suppose E is a non-empty set of real numbers which is bounded above, then E has a supremum in $\mathbb{R}$. Suppose E is a non-empty set of real numbers which is bounded below, then E has a infimum in $\mathbb{R}$.

Theorem of Limit of Monotone Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists. Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Nested Intervals Theorem If the interval $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\}$ and $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

Real Number Space $\mathbb{R}$

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset E of the real line; then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Propositions of Continuous Function on Closed Interval

Theorem on Boundedness If function f is continuous on a bounded closed interval, then it is bounded.

Hint: Bolzano-Weierstrass Theorem

Hint: Heine - Borel Theorem on Finite Coverings

Theorem on Maximum and Minimum Values If function f is continuous on a bounded closed interval, then it has (reaches) maximum and minimum values.

Hint: Theorem of Supremum and Infimum + Bolzano-Weierstrass Theorem

Theorem on Solution If function f is continuous on a bounded closed interval $[D, U]$, and $f(D)f(U) < 0$, then there exists a solution on (D, U) to equation $f(x) = 0$.

Hint: Cantor Nested Intervals Theorem

Propositions of Continuous Function on Closed Interval

Theorem on Inverse Function If function $y = f(x)$ is continuous and strictly monotonic increasing (or decreasing) on a bounded closed interval $[D, U]$, and $f(D) = L$ and $f(U) = B$, then there exists a continuous and strictly monotonic increasing (or decreasing) inverse function $x = g(y)$ on bounded closed interval $[L, B]$ (or $[B, L]$).

Cantor Theorem on Uniform Continuity If function f is continuous on a bounded closed interval, then it is uniformly continuous.

Hint: Bolzano-Weierstrass Theorem

Hint: Heine - Borel Theorem on Finite Coverings

Linear Space = Vector Space

A vector space over a field F is a set V together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors v_1 and v_2 and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set V).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar a and any vector v and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set V).

Elements of V are commonly called vectors. Elements of F are commonly called scalars.

Linear Space = Vector Space

(1) Commutativity: For all v_1 and v_2 in V, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in V.

(2) Associativity: For all v_1, v_2 and v_3 in V, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in V such that, for every element v in V, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each v in V, there exists an inverse element $-v$ in V such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field F for Scalar Multiplication: For each v in V, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any a_1 and a_2 in F and each v in V, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any a in F and each v_1 and v_2 in V, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any a_1 and a_2 in F and each v in V, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

Distance - Metric

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

For any two points $x^i = (x_1^i, \cdots, x_N^i)) \in \mathbb{R}^N$ for $i = 1, 2$, define the distance between the two points

$$\rho(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left(x_n^1 - x_n^2\right)^2 \right\}^{\frac{1}{2}} = \sqrt{\sum_{n=1}^{N} \left(x_n^1 - x_n^2\right)^2}.$$

Then we have the following three propositions.

(1) Positivity: $\rho(x^1, x^2) \geqslant 0$. $\rho(x^1, x^2) = 0$ if and only if $x^1 = x^2$.

(2) Symmetry: $\rho(x^1, x^2) = \rho(x^2, x^1)$.

(3) Triangle Inequality: $\rho(x^1, x^3) \leqslant \rho(x^1, x^2) + \rho(x^2, x^3)$.

From the triangle inequality, $\rho(x^1, \cdot)$ is a continuous function on x_2:

$$\lim_{\rho(x_n^2, x^2) \to 0} \rho(x^1, x_n^2) = \rho(x^1, x^2).$$

Norm

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

For any two points $x = (x_1, \cdots, x_N)) \in \mathbb{R}^N$, define the norm as

$$\|x\| = \left\{\sum_{n=1}^{N} x_n^2\right\}^{\frac{1}{2}} = \sqrt{\sum_{n=1}^{N} x_n^2} = \rho(x, 0).$$

Then we have the following three propositions.

(1) Positivity: $\|x\| \geqslant 0$. $\|x\| = 0$ if and only if $x = 0$.

(2) Symmetry: $\|x\| = \| - x\|$.

(3) Triangle Inequality: $\|x^1 + x^2\| \leqslant \|x^1\| + \|x^2\|$.

Theorem on Minimum and Maximum

Theorem on Minimum and Maximum Suppose E is a non-empty bounded set in $\mathbb{R}^N$. Then E has minimum and maximum members.

Pre-ordering: Functional Analysis

Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Ordered Set: Mathematical Analysis

Cantor Nested Intervals Theorem

Nested Intervals Theorem If the interval $\{I_n : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals in $\mathbb{R}^N$ and $\lim_{n\to\infty} |I_n| = 0$, then there exists a point $\xi = \lim_{n\to\infty} I_n \in \bigcap_{n=1}^{\infty} I_n$.

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}^N$ has a convergent subsequence.

Bolzano Theorem Each bounded sequence in $\mathbb{R}^N$ has a limit point.

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}^N$ has an accumulation point.

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}^N$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number K such that, when $n_1 > K$ and $n_2 > K$, $\|x_{n_1} - x_{n_2}\| < \varepsilon$.

Completeness: Functional Analysis

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset E of $\mathbb{R}^N$; then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Compactness: Functional Analysis